UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Second Semester – Fall 2023

Roll No. ....

Paper: Statistics-II | Course Code: STAT-103 | Time: 3 Hrs. Marks:

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. **(15x2=30)**

i) What is meant by a permutation?
ii) Define sample space.
iii) If $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \cap B) = 0.15$, find $P(A|\bar{B})$
iv) Define mutually exclusive events, give example.
v) State multiplication law of probability for two independent events A and B.
vi) If A and B are mutually exclusive events with $P(A) = \frac{2}{5}$ and $P(B) = \frac{1}{2}$, find $P(\bar{A} \cap \bar{B})$
vii) Describe the difference between a discrete and a continuous random variable.
viii) What do you mean by expected value of a random variable?
ix) For what value of k the function $f(x) = kx^3(1-x)$ for $0 \le x \le 1$ is a valid pdf?
x) Derive the mean of the binomial distribution.
xi) For a binomial distribution, the mean is 6 and the variance is 4. Find p and n.
xii) Under what conditions Poisson distribution is a limiting case of binomial distribution?
xiii) Given that X has a Poisson distribution with $P(X = 0) = 0.2$, find $P(X = 1)$
xiv) In a normal distribution lower and upper quartiles are 28 and 55 respectively, find mean and standard deviation.
xv) Write a short note on standard normal distribution.

Answer the following question. **(6x5=30)**

Q.2 The probability that a married man watches a certain television show is 0.4 and the probability that his wife watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find the probability that
(i) a married couple watches the show;
(ii) at least one person of a married couple will watch the show.

Q.3 An employer wishes to hire three people from a group of 15 applicants, 8 men and 7 women, all of whom are equally qualified to fill the position. If he selects the three at random, what is the probability that (i) all three will be men; (ii) at least two will be women?

Q.4 If on the average rain falls on twelve days in every thirty, find the probability that (i) the first three days of a given week will be fine and the remaining wet (ii) rain will fall on just three days of a given week.

Q.5 Derive the Poisson distribution as the limiting form of the binomial distribution

Q.6 A continuous random variable X has the probability density function
$f(x) = 20\,x^3\,(1-x)$ for $0 \le x \le 1$ and zero elsewhere
Find distribution function of X. Hence or otherwise find $P\left(\frac{1}{4} < X < \frac{1}{2}\right)$

Q.7 The lifetime of a certain type of battery can be closely approximated by the normal curve with a mean of 350 hours and a standard deviation of 50 hours.
(i) What percentage of these batteries will have lifetime of more than 375 hours?
(ii) Above what value will the best ten percent of the batteries lie?

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Second Semester – Spring 2022

Paper: Statistics-II Course Code: STAT-103 Time: ...

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (15x2=30)

i) What is a random experiment?
ii) What are independent events?
iii) What is the difference between classical and empirical probability?
iv) Given $P(A) = 0.60$, $P(B) = 0.40$, and $P(A \cap B) = 0.24$, find $P(A|\bar{B})$
v) State the general addition law of probability for two events A and B.
vi) What is discrete probability distribution?
vii) Show that $E[X - E(X)]^2 = E(X^2) - [E(X)]^2$
viii) Write two conditions that must be satisfied by a probability density function.
ix) Two independent random variables are such that $\text{Var}(X_1) = k$, $\text{Var}(X_2) = 2$, and $\text{Var}(3X_2 - X_1) = 25$. Find k.
x) Describe the significance of moments in a probability distribution.
xi) What is main difference between binomial and hypergeometric experiment?
xii) If mgf of X is given by $M_0(t) = (0.4 + 0.6\,e^t)^8$. Find $E(X)$ and $Var(X)$.
xiii) If X has binomial distribution with mean = 12 and variance = 4, find p and n.
xiv) Given that X has a Poisson distribution with $P(X=1) = P(X=2)$. Find $\text{Var}(X)$
xv) Write a short note on importance of the normal distribution.

Answer the following question. (6x5=30)

Q.2 The probability that a married man watches a certain television show is 0.4 and the probability that his wife watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find the probability that
(i) a married couple watches the show;
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$f(x) = 20\,x^3\,(1 - x)$ for $0 \le x \le 1$ and zero elsewhere
Find distribution function of X. Hence or otherwise find $P\left(\frac{1}{4} < X < \frac{1}{2}\right)$

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UNIVERSITY OF THE PUNJAB

Second Semester - 2018

Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Statistics-II **TIME ALLOWED: 2 Hrs. & 45 Mints.**

Course Code: STAT-103, STT-12314 Part – II **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

Q.2 Define the following: 2 x 10 = **(20)**

- (i) Simple event
- (ii) Addition Law
- (iii) Probability distribution function
- (iv) Mathematical expectation
- (v) Moment generating function
- (vi) Binomial distribution
- (vii) Multinomial distribution
- (viii) Cumulative distribution function
- (ix) Mean and variance of Hypergeometric distribution
- (x) M.G.F. of Normal distribution

Q.3 The probability that an automobile being filled with gasoline will also need an oil change is 0.35; the probability that it needs a new oil filter is 0.45; and the probability that both the oil and filter need changing is 0.15. **(07)**

(a) if the oil had to be changed, what is the probability that a new oil filter is needed?

(b) What is the probability that at least one of these need change?

Q.4 Derive the variance of the hypergeometric distribution. **(08)**

Q.5 Derive upper and lower quartiles and the quartile deviation of the Normal distribution. **(08)**

Q.6 If X is b (x; 20, 0.45) Find P (X = 12). Then find the approximation to this probability using **(07)**

(a) the Poisson distribution

(b) the normal distribution

UNIVERSITY OF THE PUNJAB

Second Semester - 2018

Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Statistics-II **TIME ALLOWED: 15 Mints.**

Course Code: STAT-103, STT-12314 Part – I (Compulsory) **MAX. MARKS: 10**

Attempt this Paper on this Question Sheet only.

Please encircle the correct option. Each MCQ carries 1 Mark. This Paper will be collected back after expiry of time limit mentioned above.

Q.1 Encircle the correct answer in the following. **(10)**

I. A compound event includes
(a) at least four outcomes (b) one and only one outcome
(c) at least two outcomes (d) all the outcomes of an experiment

II. Which of the following values cannot be the probability of an event?
(a) .82 (b) 0
(c) 1.75 (d) 0.36

III. The two events A and B are mutually exclusive. Which one of the following statements must be true?
(a) $P(A \cap B) = 0$. (b) $P(A \cap B) = 1$
(c) $P(A \cup B) = 0$. (d) $P(A \cup B) = 1$.

IV. If X and Y are two independent random variables, then $Var(X - Y)$ is equal to
(a) $Var(X) - Var(Y)$ (b) $Var(X) + Var(Y)$
(c) $Var(X) + Var(Y) - 2\,COV(X,Y)$ (d) None of above

V. If X and Y are two random variables, then E(X + Y) is equal to
(a) $E(X) + E(Y)$ (b) $E(X) + Y$
(c) $E(X) - E(Y)$ (d) None of above

VI. For a binomial distribution, the mean and variance are related by:
(a) $\mu < \sigma^2$ (b) $\mu = \sigma^2$
(c) $\mu > \sigma^2$ (d) $\mu < \sqrt{\sigma^2}$

VII. For a Poisson distribution, the mean and variance are related by:
(a) $\mu = \sigma^2$ (b) $\mu < \sigma^2$
(c) $\mu > \sigma^2$ (d) None of above

VIII. A binomial distribution may be approximated by a Poisson distribution when
(a) n is large and p is small (b) n is small and p is large
(c) n is small and p is small (d) n is large and p is large

IX. The middle area under the normal curve with $\mu \pm 2\sigma$ is
(a) 0.6827 (b) 1.0000
(c) 0.9545 (d) 0.9973

X. In a normal distribution, mean deviation is equal to
(a) σ (b) 0.8σ
(c) 0.6745σ (d) 2.0σ

UNIVERSITY OF THE PUNJAB

Second Semester - 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Statistics-II
Course Code: STAT-103, STT-12314

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions. Use of Scientific Calculators and Statistical tables is allowed but exchange of anything i.e. calculators etc. is not allowed.

Section-II

Q.2 Define the following: **(20)**

(i) Compound event
(ii) Probability density function
(iii) Mathematical Expectation
(iv) Binomial distribution
(v) Bay's Theorem
(vi) Harmonic Mean for probability distribution
(vii) Multiplication Law of Probability
(viii) Conditional Probability distribution
(ix) Cumulative distribution function for discrete variable
(x) Area between $\mu-\sigma$ *and* $\mu+\sigma$ for Normal curve.

Q.3 The probability that an industry will locate in city A is 0.7, the probability that it will locate in city B is 0.4, and the probability that it will locate in either A or B or both is 0.8. What is the probability that the industry will locate **(06)**
(a) in both cities
(b) in neither city

Q.4 A random variable X has the probability density function **(06)**

$$f(x)=\begin{cases} x & \text{for } 0<x<1 \\ 2-x & \text{for } 1\le x<2 \end{cases}$$

Find P (0 < X < 2) and P (X < 1.5)

Q.5 From a lot of 15 missiles, 6 are selected at random and fired. If the lot contains 4 defective missiles that will not fire, what is the probability that **(06)**
(a) All 6 will fire?
(b) At most one will not fire?

Q.6 Derive the variance of Poisson distribution. **(06)**

Q.7 The I.Q. of 1000 applicants to a certain college are approximately normally distributed with a mean of 110 and a standard deviation of 15. If the college requires an IQ of at least 90, how many of the applicants will be rejected on this basis regardless of their other qualifications? **(06)**

UNIVERSITY OF THE PUNJAB

Roll No. ....

Second Semester - 2017

Examination: B.S. 4 Years Programme

PAPER: Statistics-II
Course Code: STAT-103, STT-12314

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

Note: Attempt all questions. Use of Scientific Calculators and Statistical tables is allowed but exchange of anything i.e. calculators etc. is not allowed.

Section-I

Q.1 Encircle the correct answer in the following. **(10)**

I. P (A) = 0.6, P (B) = 0.5, which of the following statement is true?
(a) A and B are mutually exclusive (b) A and B are not mutually exclusive
(c) A and B are independent (d) A and B are dependent

II. If X is a random variable and a and b are constant then Var (aX + b) is equal to
(a) $a^2 Var(X)+b$ (b) a Var (X)
(c) $a^2 Var(X)$ (d) $a\,Var(X)+b$

III. The number of ways in which four looks can be arranged on a shelf is
(a) 4 (b) 6
(c) 24 (d) 12

IV. The parameters of the probability distribution $p(x)=\overset{n}{\underset{x}{C}}\,p^x q^{n-x}$, such that p + q = 1, are
(a) x and n (b) x and p
(c) x and q (d) n and (p or q)

V. In a Poisson distribution
(a) The mean and variance are equal (b) The mean and S.D. are equal
(c) The mean is greater than the variance (d) None of these

VI. In a normal distribution $N(\mu,\sigma^2)$ the semi-inter quartile range is equal to
(a) 0.7979 σ (b) 0.6745 σ
(c) 0.9544 σ (d) 0.9973 σ

VII. Mean of the distribution, $f(x)=\frac{1}{5};\ 5\le X\le 10,$ is
(a) 25 (b) 7.5 (c) 2.5 (d) 3.75

VIII. If X is a discrete random variable, then the function f(x) is
(a) A probability density function (b) A probability distribution function
(c) A cumulative distribution function (d) None of these

IX. The mean of the uniform distribution $f(x)=\frac{1}{n},\ x=1,2,3,\dots\dots\dots,n$ is
(a) $\frac{n-1}{2}$ (b) $\frac{n+1}{2}$ (c) $\frac{n^2-1}{12}$ (d) $\frac{n^2+1}{12}$

X. The normal distribution will be less spread out when
(a) The mean is small (b) The median is small
(c) The mode is small (d) The S.D. is small

UNIVERSITY OF THE PUNJAB

Second Semester 2015
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Statistics-II
Course Code: STAT-103,

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2. Briefly answer the following questions **(2x10)**

i) Write the properties of random experiment.

ii) Define the marginal probability.

iii) Define dependent and independent events with examples.

iv) Define the terms: (i) event (ii) sure event.

v) What are its three properties of random experiment?

vi) Write the properties of probability density function.

vii) Write the properties of hyper geometric experiment.

viii) Define the normal distribution.

ix) Write any four properties of normal distribution.

x) Let a random variable Z follow the standard normal distribution then find $P(0.6 \le z \le 1.67)$ and also $P(-1.67 \le z \le -0.6)$.

Q.3.a) Prove the addition law of Probability for any two events A & B. **(10)**

b) The letters of the word 'TRIANGLE' are scrambled and letters are rearranged. Find the probability that a word will (i) begin with T (ii) begin with t or E (iii) begin with T and end with E (iv) T and E are next to each other.

Q.4.a) A random variable 'X' is of continuous type with p.d.f. **(10)**

f(x) = 2x   0<x<1

f(x) = 0   otherwise

Find $(i)\ p(x \le 1/2)$ $(ii)\ p(x > 1/4)$ $(iii)\ p\left(x = \frac{1}{2}\right)$ $(iv)\ p\left(1/4 \le x < \frac{1}{2}\right)$

b) A man tosses two fair dice. What is the conditional probability that the sum of two dice will be 7, given that (i) the sum is odd (ii) the sum is greater than 6 (iii) the two dice has the same outcomes.

Q.5. A lawyer commutes daily from his sub urban home to his mid-town office. On average the trip one way takes 24 minutes with a standard deviation of 3.8 minutes. Assume the distribution of the trip times to be normally distributed. **(10)**

a. What is the probability that a trip will take at least $\frac{1}{2}$ hour?

b. If the office opens at 9:00 am and he leaves his house at 8:45 am daily. What percentage of the time is he late for work?

c. If he leaves the house at 8:35 am and coffee is served at the office from 8:50 am until 9:00 am, what is the probability that he misses the coffee?

UNIVERSITY OF THE PUNJAB

Roll No. ....

Second Semester 2015

Examination: B.S. 4 Years Programme

PAPER: Statistics-II
Course Code: STAT-103,

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE

Q.1. Chose the correct option. **(1x10)**

i) Which of the following values cannot be the probability of an event.

(a) 0.8 (b) 0.0 (c) 1.75 (d) 0.36

ii) In a group of 400 families, 300 own houses. If one family is randomly selected from this group the probability that this family owns a house is

(a) 0.75 (b) 0.25 (c) 0.80 (d) 0.40

iii) If X and Y are mutually exclusive events from a set of events, then if X occurs

(a) Y must also occur
(b) Y cannot occur
(c) X & Y independent
(d) None

iv) Given P(A) = 0.40, $P(A \cap B) = 0.15$, P(B) = 0.50, P(A/B) is

(a) 0.60 (b) 0.30 (c) 0.80 (d) 1.26

v) If X and Y are two random variables then E (X+Y) is

(a) E(X) + Y (b) E(X) E(Y) (c) E(X) + E(Y) (d) E(X) - E(Y)

vi) Let random variable X & Y are independent then Var (X-Y) is equal to

(a) Var(X) + var (Y)
(b) var(X) - var (Y)
(c) var(X)
(d) var(Y)

vii) For Poisson distribution

(a) $\mu > \sigma^2$ (b) $\mu \leq \sigma^2$ (c) $\mu = \sigma^2$ (d) None

viii) The binomial distribution is approximated by Poisson distribution when

(a) n is small, p is large
(b) n is large, p is small
(c) n is small
(d) p large

ix) In normal distribution, the mean deviation is

(a) 2.45 σ (b) 0.8 σ (c) 0.67 σ (d) 0.64 σ

x) In normal distribution the middle area of $(\mu + 2\sigma)$ is

(a) 0.6827 (b) 0.7979 (c) 0.9545 (d) 0.9973